Year 12 Further Maths - Momentum and collisions

Similar questions - Direct collisions of particles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Three particles A, B and C, of masses 2 kg, 3 kg and m kg respectively, lie at rest in that order on a smooth horizontal line, with B between A and C. The coefficient of restitution between any two of them is $\frac{2}{3}$. Particle A is projected towards B with speed 6 $\mathrm{m\,s^{-1}}$.

(a) Show that, after the collision between A and B, A is at rest and B has speed 4 $\mathrm{m\,s^{-1}}$. **[4]**

(b) B then collides with C. Find, in terms of m, the velocity of B after this collision. **[5]**

(c) Given that the direction of motion of B is reversed in this collision, find the range of possible values of m. **[2]**

(d) Show that, in the case where the direction of motion of B is reversed, there is a second collision between A and B. **[1]**

2 Two stones, A of mass 4 kg and B of mass 6 kg, slide towards each other along the same straight line on a horizontal sheet of ice. Stone A has speed 5 $\mathrm{m\,s^{-1}}$ and stone B has speed 2 $\mathrm{m\,s^{-1}}$. The stones collide directly and the direction of motion of A is reversed; immediately afterwards its speed is 1.2 $\mathrm{m\,s^{-1}}$.

(a) Find the speed of B immediately after the collision. **[3]**

(b) Find the coefficient of restitution for the collision. **[2]**

(c) Find the total loss of kinetic energy. **[3]**

(d) Find the magnitude of the impulse exerted by A on B. **[2]**

(e) Give a reason why the collision is not perfectly elastic. **[1]**

3 Two particles P of mass m kg and Q of mass 3 kg move towards each other along a straight line on a smooth horizontal surface. P has speed u $\mathrm{m\,s^{-1}}$ and Q has speed 1 $\mathrm{m\,s^{-1}}$. The coefficient of restitution between them is 0.75. After they collide, the direction of motion of each is reversed; P now has speed 1.3 $\mathrm{m\,s^{-1}}$ and Q has speed 3.2 $\mathrm{m\,s^{-1}}$.

(a) Find u and m. **[5]**

(b) Show that approximately 36% of the kinetic energy of the system is lost. **[3]**

Q goes on to collide directly with a third particle R of mass 4 kg that is moving towards Q with speed 1.5 $\mathrm{m\,s^{-1}}$. The coefficient of restitution between Q and R is f.

(c) Given that there are no further collisions among P, Q and R, find the range of possible values of f. **[6]**

Question 1

(a) Momentum: $2(6) = 2v_A + 3v_B$. Restitution: $v_B - v_A = \frac{2}{3}(6) = 4$. So $12 = 2v_A + 3(v_A + 4)$, giving $v_A = 0$ and $v_B = 4$

(b) Momentum: $3(4) = 3v + mw$ where v is the velocity of B and w that of C. Restitution: $w - v = \frac{2}{3}(4) = \frac{8}{3}$. So $12 = 3v + m\left(v + \frac{8}{3}\right)$ and $v = \dfrac{36 - 8m}{3(3+m)}$

(c) Reversed when $v < 0$: $36 - 8m < 0$, so $m > 4.5$

(d) A is at rest and B now moves towards A (with speed $\dfrac{8m-36}{3(3+m)}$), so they collide again

Question 2

(a) Take A's original direction as positive: $4(5) + 6(-2) = 4(-1.2) + 6v_B$, so $8 = -4.8 + 6v_B$ and $v_B = 2.13\ \text{m s}^{-1}$ (in the direction of A's original motion)

(b) $e = \dfrac{2.133 + 1.2}{5+2} = 0.476$

(c) Before: $\frac{1}{2}(4)(25) + \frac{1}{2}(6)(4) = 62$ J. After: $\frac{1}{2}(4)(1.44) + \frac{1}{2}(6)(2.1333^2) = 2.88 + 13.65 = 16.53$ J. Loss $= 45.5$ J

(d) $I = 6(2.1333 - (-2)) = 24.8$ N s

(e) $e < 1$ (or kinetic energy is lost)

Question 3

(a) Take P's original direction as positive. Restitution: $0.75(u+1) = 1.3 + 3.2 = 4.5$, so $u = 5$. Momentum: $5m - 3(1) = -1.3m + 3(3.2)$, so $6.3m = 12.6$ and $m = 2$

(b) Before: $\frac{1}{2}(2)(25) + \frac{1}{2}(3)(1) = 26.5$ J; after: $\frac{1}{2}(2)(1.69) + \frac{1}{2}(3)(10.24) = 17.05$ J. Loss $= 9.45$ J, which is $35.7\% \approx 36\%$

(c) Q moves at 3.2 (positive), R at -1.5. Momentum: $3(3.2) + 4(-1.5) = 3.6 = 3v_Q + 4v_R$; restitution: $v_R - v_Q = 4.7f$. So $v_Q = \dfrac{3.6 - 18.8f}{7}$. P moves at -1.3. Q will catch P again if $v_Q < -1.3$, so we need $\dfrac{3.6 - 18.8f}{7} \geqslant -1.3$, giving $f \leqslant \dfrac{127}{188} = 0.676$. ($R$ always moves away from Q since $v_R > v_Q$.) So $0 \leqslant f \leqslant 0.676$